

HARSHA VARDHAN UPADRASTA

AI Automation QA Engineer | GenAI & Intelligent Quality Engineering | Azure OpenAI & Copilot Studio

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PROFESSIONAL SUMMARY

AI Automation QA Engineer with 3+ years of experience in intelligent quality engineering, specialising in automated validation of web, API, backend, and LLM-powered systems. Hands-on builder of **Azure OpenAI & Copilot Studio agents** — including test-case automation agents, MongoDB query agents, Apigee API creation agents, and documentation-to-testcase generation agents. Expert in Python, Playwright, PyTest, and BDD, with a strong track record of embedding AI-driven quality gates into Azure DevOps and Jenkins CI/CD pipelines. Passionate about applying GenAI to accelerate testing cycles, reduce defect leakage, and deliver scalable quality engineering solutions.

CORE SKILLS

Quality Engineering & Automation	Playwright, PyTest, BDD (Cucumber/Gherkin), E2E Validation, Regression, Integration, Framework Architecture
Azure OpenAI & Copilot Studio	Agent Builder, Power Virtual Agents, Custom Connectors, Prompt Flow, Adaptive Cards, Bot Framework
AI Agent Development	Copilot Studio Test-Case Agent, MongoDB Query Agent, Apigee API Creation Agent, Doc-to-Testcase Agent, Web Methods Assistant
Generative AI & LLM Evaluation	LLM Validation (GPT-4, Claude, Gemini), Prompt Engineering, Model Output Evaluation, Conversational Testing
API & Service Validation	REST, SOAP, Postman, SOAPUI, Contract Testing, Apigee API Gateway, Data Integrity Checks
CI/CD & DevOps	Azure DevOps Pipelines, Jenkins, Git, GitHub, Automated Quality Gates, Parallel Execution
Cloud Platforms	AWS (Cloud Practitioner), Azure (CI/CD, OpenAI, Bot Services)
Programming & Practices	Python, Microservices Validation, Agile/Scrum, Defect RCA, JIRA, MongoDB

AI AGENT PROJECTS (AZURE OPENAI & COPILOT STUDIO)

Copilot Studio Test-Case Automation Agent

Azure OpenAI · Copilot Studio · Power Automate · PyTest · Azure DevOps

- Designed and deployed an intelligent Copilot Studio agent that auto-generates PyTest / BDD test cases from natural-language requirements and user stories.
- Integrated the agent with Azure DevOps work items — on ticket creation the agent pulls acceptance criteria, scaffolds test functions, and commits them to the repo via Power Automate flows.
- Reduced manual test-authoring effort by ~60 %, enabling same-day test coverage for new features in Agile sprints.
- Built adaptive-card based chat interface allowing QA engineers to refine generated test cases conversationally within Microsoft Teams.

MongoDB Query Agent

Azure OpenAI (GPT-4o) · Copilot Studio · MongoDB Atlas · Custom Connectors

- Built a natural-language-to-MongoDB-query agent that translates plain-English questions into optimised aggregation pipelines and find() queries.
- Implemented schema-aware prompt engineering to handle complex nested documents and array operations without manual query writing.
- Exposed the agent as a Copilot Studio topic, enabling non-technical stakeholders to query production databases safely through an approval workflow.
- Added query explain-plan analysis to flag slow queries and suggest index improvements automatically.

Apigee API Creation Agent

Azure OpenAI · Copilot Studio · Google Apigee · OpenAPI 3.0 · Power Automate

- Engineered a conversational agent that scaffolds full Apigee API proxies (OpenAPI spec, proxy bundle, policies) from a plain-English API description.
- Automated creation of security policies (OAuth 2.0, API-key verification, quota), traffic-management policies, and fault-handling rules.
- Cut API onboarding time from days to under 2 hours by generating ready-to-deploy proxy bundles pushed directly to Apigee via REST management API.
- Integrated with Azure DevOps pipelines to trigger automated contract tests immediately after proxy creation.

Documentation-to-Test-Case Generation Agent

Azure OpenAI · Copilot Studio · SharePoint · Playwright · Confluence REST API

- Built an agent that ingests technical documentation (PDF, Word, Confluence pages, SharePoint) and extracts structured test scenarios covering happy paths, edge cases, and negative flows.
- Leveraged RAG (Retrieval-Augmented Generation) over internal documentation corpus to ensure context-aware, project-specific test generation.
- Produced Gherkin feature files and Playwright automation scripts directly from documentation, achieving 45 % faster test suite bootstrapping for new projects.
- Enabled continuous sync — agent monitors document changes via webhooks and flags impacted test cases for review.

WebMethods Integration Assistant Agent

Azure OpenAI · Copilot Studio · Software AG webMethods · REST/SOAP APIs

- Developed a conversational assistant to help integration developers query webMethods IS service metadata, troubleshoot flow errors, and generate integration stubs.
- Agent parses webMethods audit logs and error journals to provide root-cause analysis suggestions in natural language, cutting incident resolution time by 35 %.
- Supports on-demand generation of SOAP/REST adapter configurations and trigger definitions through guided dialogue.

PROFESSIONAL EXPERIENCE

Assistant Manager – AI Automation QA

Sep 2025 – Present

VOIS (Vodafone Intelligent Solutions)

- Architected intelligent automation frameworks using Python, PyTest, Playwright, and BDD, improving maintainability by 40%.
- Deployed Azure OpenAI & Copilot Studio agents for automated test generation, defect prediction, and documentation-driven test case creation.
- Developed scalable validation suites for UI, API, and backend systems, increasing regression coverage by 55%.
- Automated REST and SOAP service validation using Postman and SOAPUI, reducing production defects by 30%.
- Integrated automated quality checks into Azure DevOps and Jenkins pipelines, cutting release validation time by 45%.
- Evaluated LLM-powered features by validating prompt behaviour, response accuracy, and conversational consistency.
- Led GenAI PoCs for AI-assisted test generation, intelligent defect prediction, and Copilot Studio agent development.
- Optimised execution via parallelisation and analytics-driven reporting, improving pipeline efficiency by 35%.
- Collaborated with engineering teams to reduce defect leakage by 25% through structured root-cause analysis.

Senior Executive – Automation QA Engineer

Aug 2023 – Sep 2025

VOIS (Vodafone Intelligent Solutions)

- Designed and maintained modular Python-based automation frameworks using PyTest and BDD.
- Automated UI and service-level validation for microservices-based applications.
- Embedded validation workflows into Jenkins CI pipelines to enable continuous quality checks.
- Executed functional, regression, and integration assurance in Agile environments.
- Began early exploration of LLM-powered test generation tooling using OpenAI APIs.

Pioneer Technical Graduate – QA Automation

Aug 2022 – Aug 2023

VOIS (Vodafone Intelligent Solutions)

- Built automated verification scripts using Python, PyTest, and BDD frameworks.
- Supported regression assurance and API-level validation initiatives.
- Enabled CI-based execution for faster feedback cycles.

EDUCATION

M.Tech – Computer Science & Engineering (AI & ML)

2025

Jawaharlal Nehru Technological University, Kakinada

B.Tech – Computer Science & Engineering

2018 – 2022

VSM College of Engineering

CERTIFICATIONS & AWARDS

- AWS Well-Architected Proficient – Amazon Web Services
- Automation Pro I, II, III – Workato
- Generative AI Foundations – Skillssoft / MIT Horizon (Critical Emerging Technologies)
- DevSecOps Foundation – Vodafone
- Cyber Security Essentials – Vodafone
- **STAR Award – VOIS** (Multiple Times) — Recognised for outstanding contributions to AI-driven quality engineering initiatives